Research Article
Liver Transplantation

JOURNAL OF HEPATOLOGY

Variation in intention-to-treat survival by MELD subtypes: All models created for end-stage liver disease are not equal

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Graphical abstract

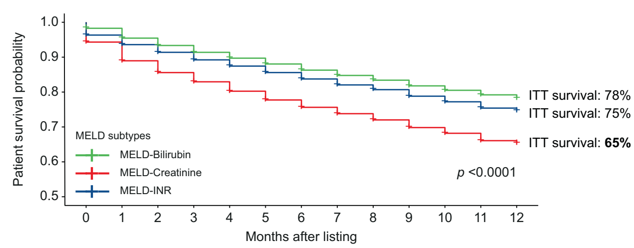
Hypothesis: *Identical MELD score values at listing, outcomes before and after liver transplantation may vary if the predominant driver of the MELD score is serum creatinine versus serum bilirubin or INR*

Methods

- Adult patients listed for liver transplant between 2016-2020
- Exclusion: Patients receiving -
  - MELD exceptions
  - Dual organ transplantation
- Primary outcome: Intent-to-treat survival (within 1 year from listing)

Results

- ITT survival lower for each MELD subtype for **females** as compared to males
- **MELD-Cr** was associated with higher waitlist mortality and lower liver transplant rates



Conclusion

For equivalent listing practices, registrants with MELD-Cr subtype have lower ITT survival

Highlights

- Patients with MELD scores dominated by Cr have lower 1 year ITT survival compared to scores dominated by Br or INR
- For each of the subtypes, females have lower 1 year ITT survival compared to males
- Within the MELD-Cr subtype, those with AKI were less likely to get transplanted and more likely to die than those with CKD

Highlights

- Patients with MELD scores dominated by Cr have lower 1 year ITT survival compared to scores dominated by bilirubin or INR.
- For each of the subtypes, females have lower 1 year ITT survival compared to males.
- Within the MELD-Cr subtype, those with AKI were less likely to get transplanted and more likely to die than those with CKD.

Impact and implications

The model for end-stage liver disease (MELD) score is an excellent predictor of waitlist mortality; however, our work highlights that the driver of a patient's MELD score matters and particularly those driven by elevated creatinine are associated with lower 1-year intent-to-treat survival. The 1-year intent-to-treat survival is also lower for women compared to men within the Cr-dominant subtype. These results are important for physicians and patients undergoing LT evaluation as creatinine may serve as a marker of prognosis and even if creatinine levels improve the prognosis remains poor, necessitating discussion about alternative pathways for transplant. Our work also highlights that the type of kidney injury matters, in that those with acute kidney injury were more likely to die or remain on the waitlist than those with chronic kidney disease within the creatinine-dominant subtype.

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Variation in intention-to-treat survival by MELD subtypes: All models created for end-stage liver disease are not equal

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Background & Aims: Kidney dysfunction is a major determinant of prognosis in patients with decompensated cirrhosis awaiting transplantation. We hypothesized that for identical model for end-stage liver disease (MELD) scores at listing, outcomes before and after liver transplantation may vary if the predominant driver of the MELD score is serum creatinine (Cr) *vs.* serum bilirubin (Br) or international normalized ratio (INR).

Methods: We evaluated all adult patients registered for liver transplantation (LT) between 2016–2020 and excluded patients receiving MELD exceptions or undergoing dual organ transplantation. Using K-Means clustering analysis, we classified each patient as MELD-Br, MELD-INR or MELD-Cr depending on the dominant variable for their MELD score. The primary outcome was intent-to-treat (ITT) survival, defined as survival within 1 year from listing with or without LT.

Results: MELD scores of LT waitlist registrants were clustered into three subtypes: MELD-Br (n = 13,658), MELD-INR (n = 13,809), and MELD-Cr (n = 12,412). One-year ITT survival rates were 78% (MELD-Br), 75% (MELD-INR), and 65% (MELD-Cr), p <0.01. ITT survival was lower for each MELD subtype for females compared to males (*e.g.* MELD-Cr: 63% females *vs.* 67% males, p <0.0001). The MELD-Cr subtype had the highest MELD at listing (MELD-Cr 23.4 *vs.* MELD-Br 19.2 *vs.* MELD-INR 21.0) and the largest decline in MELD over 3 months (23% *vs.* 12% *vs.* 21%). In adjusted analyses including MELD-Na, MELD-Cr was associated with higher waitlist mortality (hazard ratio 1.339, 95% CI 1.279-1.402) and lower LT rates (hazard ratio 0.688, 95% CI 0.664-0.713) compaed to the other subtypes.

Conclusions: For equivalent listing practices, registrants with the MELD-Cr subtype have lower ITT survival. MELD subtype may serve as a more sophisticated variable for dynamic assessment of mortality risk and to guide organ allocation.

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Introduction

The model for end-stage liver disease (MELD) score (and its subsequent iterations-MELD-Na and MELD 3.0) is an excellent predictor of waitlist mortality and is the foundation of urgency-based organ allocation with or without the addition of serum sodium.[1,2] However, there are several reasons why the MELD score may incompletely capture prognosis after listing across all registrants and diagnoses. First, in the current era, characteristics of registrants awaiting liver transplant (LT) have changed.[3] Specifically, registrants are older and the proportion of patients with chronic kidney disease (CKD) has increased from 7.8% to 14.6%.[4] Second, sex-based disparities in LT rates persist and may have increased in the MELD-Na era.[5,6] It is currently unclear how much of the impact is abrogated by the introduction of MELD 3.0.[7] Third, for a given MELD score, drivers of elevated MELD scores (*e.g.* creatinine, international normalized ratio [INR], or bilirubin) may have a variable impact on prognosis, waitlist mortality, and transplant rates. For a given MELD and serum creatinine, prognosis may differ by the presence of acute kidney injury (AKI) or CKD.[8–10] Alternatively, for a given MELD score, elevated serum bilirubin driven by chronic elevation in patients with cholestatic liver disease may be associated with a better outcome than a similar acute level of bilirubin that is acutely elevated in patients with acute-on-chronic liver failure.[11] Hence, even for equivalent MELD scores, outcomes may differ by the presence of underlying comorbidities, underlying drivers of MELD score, and gender.

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Assessing the impact of MELD subtypes and outcomes on the waitlist is important for several reasons. First, the impact of serum creatinine on assessment of liver-related mortality remains important. There is conflicting data regarding whether, for a given MELD score, creatinine-predominant MELD scores are associated with better or worse outcomes than those with normal creatinine.[12,13] Second, understanding drivers of waitlist mortality and LT outcomes by MELD subtype may help identify the underlying disparities noted in waitlisting among women.[14] Finally, if outcomes differ by predominant MELD subtype from the time of listing (intention-to-treat [ITT] survival), examination of patterns may allow for more informed discussions with patients regarding priority, prognosis, and organ allocation policy, especially in an era where treatments are available for hepatorenal syndrome (HRS)-AKI that decrease serum creatinine without significantly altering mortality. We hypothesized that for the same MELD score at listing, relevant clinical outcomes before and after LT vary by MELD subtype, and are worse among patients with a creatinine-dominant subtype.

Patients and methods

The aim of the study was to identify unique subsets of registrants on the waitlist with similar MELD scores that are bilirubin dominant (MELD-Br), creatinine dominant (MELD-Cr), and INR dominant (MELD-INR). The methodology used to characterize "dominance" is described below. The primary outcome was to assess differences by ITT survival from time of registration.[15] The secondary outcomes looked at specific components of waitlist mortality, LT rates, and differences by sex.

Data source

This study used data from the Scientific Registry of Transplant Recipients (SRTR). The SRTR data system includes data on all donor, wait-listed candidates, and transplant recipients in the US submitted by the members of the OPTN (Organ Procurement and Transplantation Network). The Health Resources and Services Administration, US Department of Health and Human Services provides oversight on the activities of the OPTN and SRTR contractors.

Case ascertainment

The study was approved by the institutional review board as exempt from human subject review. We used data submitted to the SRTR on all adult (age ≥18 years) registrants listed for primary LT between 2016–2020 in the US. We excluded patients who were listed for re-transplantation, multiple organ transplantation (including simultaneous liver-kidney transplant [SLKT] at baseline), or those that were status 1 and/or listed for acute liver failure. Patients undergoing living donor LT were also excluded because they had an alternate pathway to LT with shorter waitlist times. We excluded patients listed with MELD exceptions to focus on registrants waitlisted for decompensated liver disease (Fig. S1).

MELD subtype

The goal was to identify three distinct subtypes dominated by serum bilirubin, serum creatinine, or serum INR. MELD subtype was derived from the measured lab contribution of each component of the MELD score as below:

$$\text{Creatinine MELD contribution} = 9.57 * \log(\text{Creatinine})/\text{MELD}$$

$$\text{Bilirubin MELD contribution} = 3.78 * \log(\text{Bilirubin})/\text{MELD}$$

$$\text{INR MELD contribution} = 11.2 * \log(\text{INR})/\text{MELD}$$

Creatinine, bilirubin, and INR MELD contributions were then standardized (rescaled to mean equals to zero and standard deviation equals to one) to reduce the difference of scale and variance among these three variables. In order to avoid arbitrary classification of each subtype, we *a priori* chose K-Means clustering to differentiate MELD subtypes. Then we used K-Means clustering by first determining the optimal number of clusters K using the Elbow method. The optimal K was 3. Next, we performed K-means clustering analysis using K = 3. The mechanism of this approach is based on clustering the data points into three groups in a 3-dimensional space by iteration of search for three centroids which minimize the total within cluster sum of squares (*i.e.* total withinss). Finally, for the three clusters identified, the cluster with principal contribution from creatinine was named MELD-Cr. The cluster with principal contribution from bilirubin was named MELD-Br. The cluster with principal contribution from INR was named MELD-INR.

MELD-Na and MELD 3.0 were not used given that sodium, sex, and albumin have either an interaction or apply only over a certain range of the MELD score. In addition, the contribution is not independent of the contribution of bilirubin, creatinine, and INR. However, serum sodium levels and MELD-Na were introduced as adjusting variables in separate models to assess its individual impact in the multivariable analysis. Use of either MELD 3.0 or MELD-Na did not change the results. Use of deterministic grouping rather than K-Means clustering led to similar results for MELD subtype groups (see supplementary information).

ITT survival

The primary outcome of interest was ITT survival at 1 year from registration on the waitlist. ITT survival analyzes the results based on the initial treatment assignment and not on the LT eventually received. This method estimates the risk of mortality for a given patient from the time of registration and reflects the cumulative risks on the waitlist, during transplantation, and after transplant. The start time is registration on the waitlist. The end time is the end of 1 year (365 days), during which a patient may or may not be transplanted. Deaths or dropouts (due to being sick or for other reasons) were considered as events. If a patient received a liver transplant within a year, they were still followed until the end of 365 days. This method had previously been used to evaluate outcomes from the patient perspective in those with HIV, HCV, hepatocellular carcinoma, in European populations and US cohorts.[15–20] The secondary outcome was LT rates within 90 days of waitlisting by dominant MELD subtype after accounting for the competing risk of mortality on the waitlist. Registrants were followed from time of registration to LT, removal from the waitlist, or death. Removal for being too sick to transplant, not alive 30 days after delisting, or "other" were considered as waitlist deaths.

Secondary analyses by MELD subtype

During the study, we observed differences in LT rates by MELD-Cr *vs.* other subtypes. We further explored these differences by examining additional outcomes by MELD subtype. (a) We examined longitudinal changes in waitlist MELD (delta MELD) to assess whether lower LT rates may be related to higher MELD fluctuations by MELD subtype. (b) We evaluated whether lower LT rates were associated with the phenotype of kidney injury (AKI *vs.* CKD) using previously validated definitions and KDIGO criteria.[8,21] We defined AKI as an increase in serum creatinine by ≥0.3 mg/dl or 50% in the last 7 days or fewer than 72 days of hemodialysis. CKD was defined as eGFR <60 ml/min/1.73 m^2 (using the CKD-EPI definition) at waitlisting and then eGFR <60 ml/min/1.73 m^2 at each update until LT or last day of follow-up or ≥72 days within initiation of hemodialysis. We defined AKI on CKD if both AKI and CKD definitions were met. (c) Finally, multivariable competing risk Fine-Gray regression models were used to identify drivers of waitlist mortality and LT rates after accounting for relevant recipient and transplant-related characteristics. We adjusted for relevant covariates such as age, race/ethnicity, liver disease etiology, height, weight, BMI, MELD-Na score, and ventilator status.

Statistical analysis

Kruskal-Wallis Test was used to compare continuous variables and the Chi-square test or Fisher's exact test was used to compare categorical variables. Competing risk analysis was used to assess liver transplant rates while accounting for waitlist death/mortality and dropout. We examined ITT survival by dominant MELD subtypes overall and stratified by relevant MELD categories (MELD <15, 15-24, 25-34 and 35+) and by sex. Statistical analyses were performed in statistical software R version 4.4.0.

Results

Patient characteristics

Between 2016 and 2020 there were 39,897 patients listed with decompensated cirrhosis meeting our inclusion criteria (Fig. S1). The three MELD subtype groups identified via K-Means clustering analysis were MELD-Br (n = 13,658), MELD-Cr (n = 12,412), and MELD-INR (n = 13,809) (Table 1). Compared to MELD-Cr and MELD-INR, a significantly higher proportion of patients with the MELD-Br subtype had cholestatic liver disease (16.7% *vs*. 5.8% *vs*. 3.9%). Compared to MELD-Br and MELD-INR, a higher proportion of patients with the MELD-Cr subtype were of Hispanic ethnicity (17.4% *vs.* 14.5% for MELD-Br *vs.* 16.4% for MELD-INR) and had a higher median MELD score at registration (25.0 *vs.* 17.0 for MELD-Br and 19.0 for MELD-INR). There were no clinically relevant differences by age, sex or steatotic liver disease/cryptogenic diagnoses across groups.

Variation in ITT survival

One-year ITT survival was lowest for MELD-Cr (65%) – compared to MELD-Br (78%) or MELD-INR (75%) (p <0.01) (Fig. 1). By sex, ITT survival in the cohort was lower for each of the subtypes for females compared to males (p <0.01) (Fig. 2). For example, ITT survival was 63% for MELD-Cr in females and 67% for MELD-Cr in males. This was lower than other MELD subtypes (bilirubin or INR, rates >73%). Table 2 shows the difference in ITT survival at 1 year compared to overall average ITT survival without MELD subtypes, with lower rates seen among females.

Variation in waitlist mortality and LT rates

Overall, across all MELD scores, MELD-Cr was associated with the highest waitlist mortality (12% *vs.* 6% for MELD-Br *vs.* 8% for MELD-INR). After accounting for waitlist mortality, there was variation in the cumulative incidence of LT within 90 days by MELD subtype (Table 3). As an example for MELD 25-34, the LT rate was reduced by at least 27% for MELD-Cr (50% MELD-Cr *vs.* 77% MELD-Br and 80% MELD-INR) (Fig. 3)

Post-LT survival: LT rates were 7-10% lower for females compared to males across all MELD subtypes (Fig. S3) This difference was more pronounced when looking at MELD strata.

On adjusted analysis (Table 4), MELD-Cr was associated with higher waitlist mortality (hazard ratio 1.34, 95% CI 1.28-1.40) and lower LT rates (hazard ratio 0.69, 95% CI 0.66-0.71) compared to the other subtypes.

Change in MELD over time

We examined potential drivers of the patterns seen in the MELD-Cr group. We examined whether patients with MELD-Cr may have had rapid improvement in their MELD score after listing. MELD-Cr subtype had the highest MELD at listing (MELD-Cr 23.4 *vs.* MELD-Br 19.2 *vs.* MELD-INR 21.0) and the largest decline in serial MELD over 3 months (23% *vs.* 12% *vs.* 21%) (Fig. 4).

Kidney dysfunction phenotype

We examined whether patterns of kidney injury impacted ITT survival. A large number of patients with AKI either died or remained on the waitlist *vs.* receiving LT compared to those with AKI on CKD or CKD (Fig. 5). AKI was associated with lower rates of transplant (59.4%) compared to CKD (68.5%) or AKI on CKD (69.9%) in the MELD-Cr subtype. Specifically, for the AKI phenotype in the MELD 25-34 group, LT rates were lower (71.6% AKI *vs.* 80.6% CKD) and deaths were higher (25.3% AKI *vs*. 15.9% CKD) (Fig. S4).

Exploratory analysis of SLKT after listing for LT alone

Overall, 750 patients were initially listed for LT alone but, due to progression, were subsequently listed for SLKT. The predominant phenotype that got transplanted for SLKT in this subset were patients with AKI superimposed on CKD, suggesting that an acute deterioration in patients with underlying CKD while waiting for LT led to listing for SLKT (66.7% AKI on CKD *vs.* 54.0% CKD).

Discussion

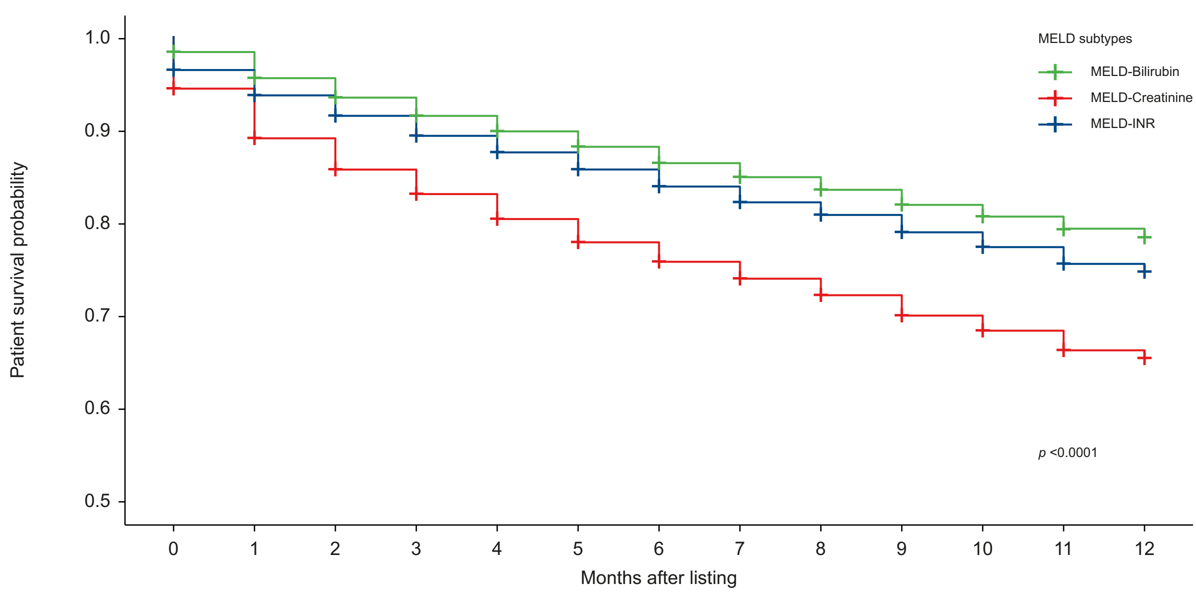
The MELD score has served as the reference standard for assessment of liver disease severity and urgency-based organ allocation decisions worldwide over more than two decades.

Table 1. Baseline characteristics of candidates at the time of registration by MELD subtypes from K-means clustering analysis.

	Bilirubin (n = 13,658)	Creatinine (n = 12,412)	INR (n = 13,809)
Age (years)	54.0 [46.0;61.0]	58.0 [50.0;64.0]	55.0 [47.0;62.0]
Female	5,762 (42.2%)	4,756 (38.3%)	5,525 (40.0%)
BMI (kg/m^2)	28.0 [24.0;32.0]	29.0 [25.0;33.0]	29.0 [25.0;33.0]
Race			
White	12,100 (88.6%)	10,930 (88.1%)	12,064 (87.4%)
Black	868 (6.4%)	911 (7.3%)	964 (7.0%)
Other	690 (5.1%)	571 (4.6%)	781 (5.7%)
Hispanic ethnicity	1,986 (14.5%)	2,155 (17.4%)	2,260 (16.4%)
Creatinine (mg/dl)	1.0 [1.0;1.1]	4.0 [1.6;4.0]	1.0 [1.0;1.2]
Bilirubin (mg/dl)	5.4 [3.2;12.2]	2.7 [1.2;9.5]	3.1 [1.7;7.5]
INR	1.4 [1.3;1.7]	1.5 [1.2;2.0]	1.9 [1.5;2.8]
Sodium (mEq/L)	136 [132;137]	136 [133;137]	136 [132;137]
MELD	17.0 [14.0;23.0]	25.0 [15.0;36.0]	19.0 [14.0;27.0]
MELD-Na	20.0 [15.0;26.0]	27.0 [17.0;37.0]	21.0 [15.0;29.0]
MELD 3.0	21.0 [16.0;26.0]	27.0 [18.0;37.0]	21.0 [16.0;30.0]
Diagnosis			
ALD	4,864 (35.6%)	4,532 (36.5%)	5,189 (37.6%)
Cholestatic	2,285 (16.7%)	716 (5.8%)	537 (3.9%)
HCV	1,220 (8.9%)	1,432 (11.5%)	1,667 (12.1%)
NASH/cryptogenic	3,079 (22.6%)	3,755 (30.3%)	3,631 (26.3%)
Others	2,202 (16.1%)	1,969 (15.9%)	2,784 (20.2%)
MELD range categories:			
1. MELD <15	3,868 (28.3%)	2,741 (22.1%)	3,941 (28.5%)
2. 15≤ MELD <25	6,986 (51.1%)	3,340 (26.9%)	5,580 (40.4%)
3. 25≤ MELD <35	2,259 (16.5%)	2,620 (21.1%)	2,669 (19.3%)
4. MELD ≥35	545 (4.0%)	3,711 (29.9%)	1,619 (11.7%)

ALD, alcohol-related liver disease; INR, international normalized ratio; MELD, model for end-stage liver disease; NASH, non-alcoholic steatohepatitis.
Median [Q1;Q3] for continuous variables. Count (percentage) for categorical variables. *p* values <0.01 for all comparisons.

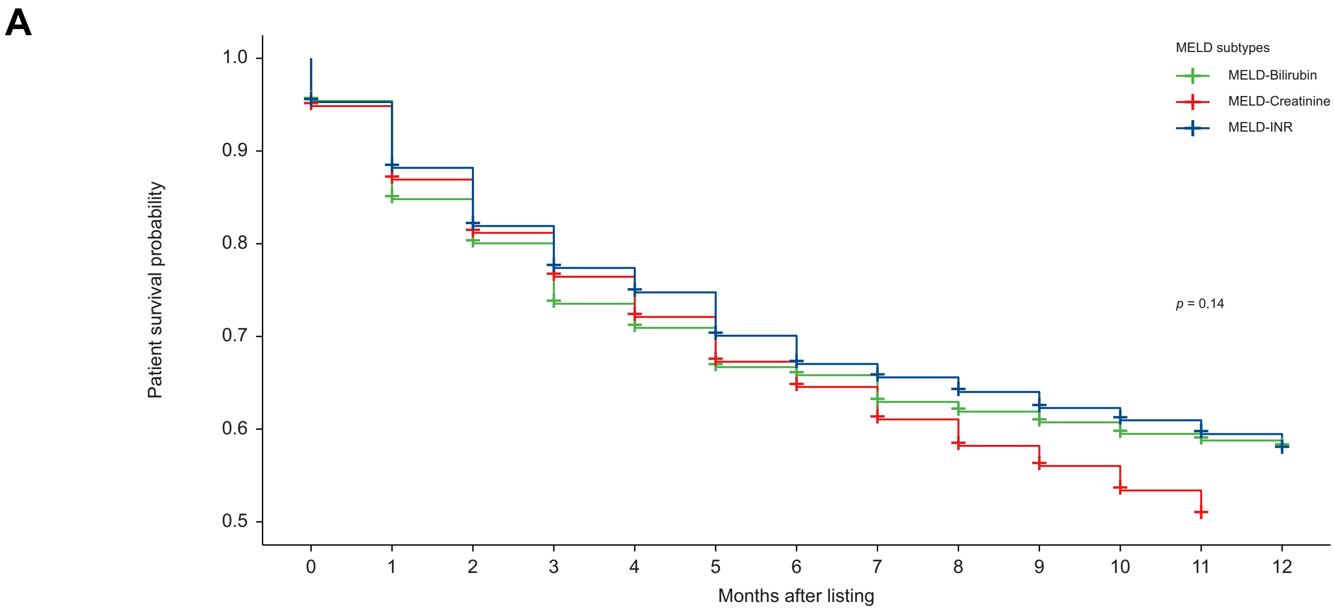
Though several studies have examined whether augmenting the score with additional variables improves prediction, detailed exploration of its components has been less frequently examined. In this study using K-Means clustering, we show that the dominant driver of the MELD score matters, specifically serum creatinine. First, among patients with equivalent MELD scores, those with scores driven by serum creatinine (MELD-Cr) have lower ITT survival. Some of the differences are explained by patient disease severity and fluctuations in the score, and the presence of certain kidney patterns. However, even after



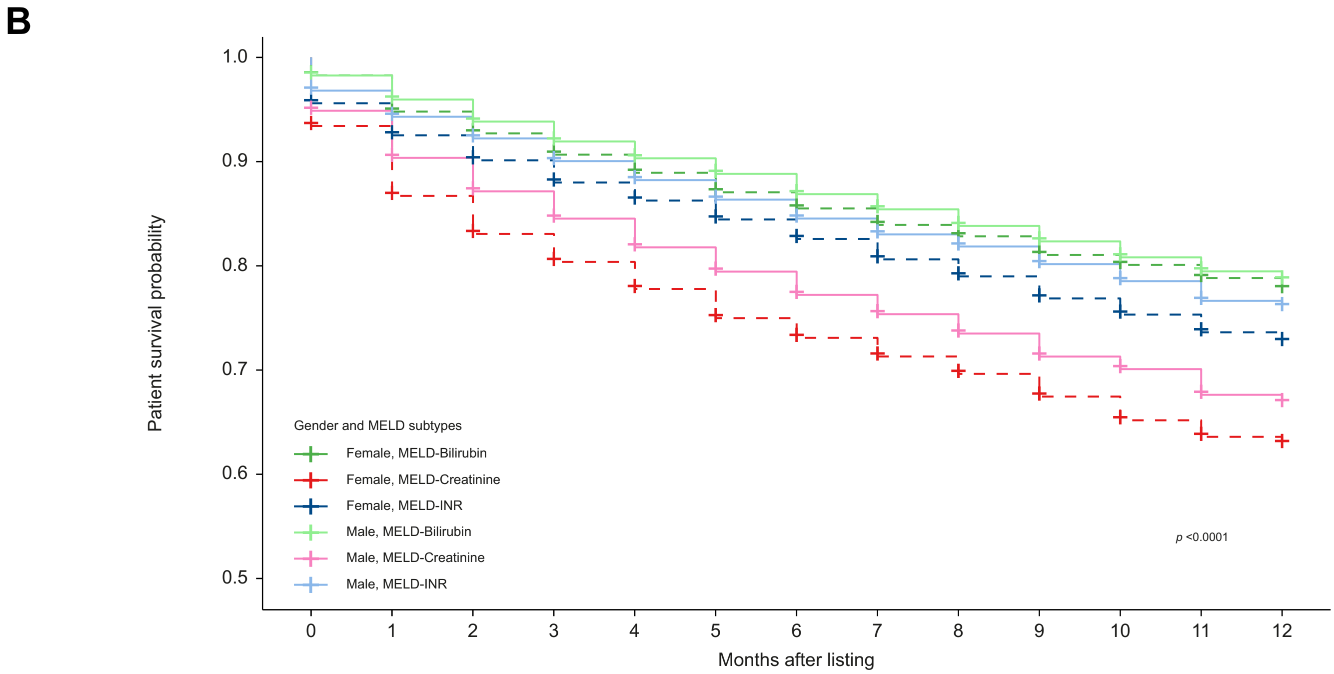
N° at risk													
MELD-Bilirubin	13,658	10,732	8,716	7,585	6,722	6,034	5,484	5,015	4,595	4,262	3,936	3,637	3,380
MELD-Creatinine	12,412	8,061	6,371	5,471	4,862	4,344	3,919	3,563	3,267	3,014	2,782	2,577	2,371
MELD-INR	13,809	9,722	7,796	6,725	5,946	5,392	4,885	4,476	4,158	3,877	3,594	3,327	3,051

Fig. 1. One-year intention-to-treat survival by MELD subtype. INR, international normalized ratio; MELD, model for end-stage liver disease. A log-rank test was used to compare the patient survival probability among MELD subtype groups. (This figure appears in color on the web.)

Variation in ITT survival by MELD subtypes



N° at risk													
MELD-Bilirubin	2,259	928	427	295	226	184	154	137	120	107	98	83	79
MELD-Creatinine	2,620	1,723	1,225	978	810	687	593	516	450	402	360	322	283
MELD-INR	2,669	1,115	534	344	264	208	161	139	125	111	94	82	70



N° at risk													
Female, MELD-Bilirubin	5,762	4,687	3,860	3,401	3,022	2,712	2,471	2,269	2,078	1,941	1,782	1,655	1,551
Female, MELD-Creatinine	4,756	3,064	2,375	2,042	1,825	1,642	1,465	1,350	1,239	1,151	1,067	985	923
Female, MELD-INR	5,525	3,855	3,161	2,751	2,439	2,235	2,029	1,860	1,728	1,605	1,484	1,376	1,259
Male, MELD-Bilirubin	7,896	6,045	4,856	4,184	3,700	3,322	3,013	2,746	2,517	2,321	2,154	1,982	1,829
Male, MELD-Creatinine	7,656	4,997	3,996	3,429	3,037	2,702	2,454	2,213	2,028	1,863	1,715	1,592	1,448
Male, MELD-INR	8,284	5,867	4,635	3,974	3,507	3,157	2,856	2,616	2,430	2,272	2,110	1,951	1,792

Fig. 2. One-year intention-to-treat survival by MELD subtype and gender. INR, international normalized ratio; MELD, model for end-stage liver disease. Log-rank tests were used to compare the patient survival probability among various groups. (This figure appears in color on the web.)

accounting for such factors, MELD-Cr was still associated with lower LT rates in relevant subsets. Though one may surmise that mortality among the different subtypes is similar on the waitlist, ITT survival analysis suggests otherwise. From time of registration, MELD-Cr patients have lower ITT survival. This difference is further aggravated for women among all subsets.

Table 2. Difference in ITT survival at 1 year compared to average survival without MELD subtypes.

Change in ITT	All MELD ranges	MELD <15	15≤ MELD <25	25≤ MELD <35	MELD ≥35
MELD subtypes					
Bilirubin	0.05	0.02	0.00	0.06	NA
Creatinine	-0.08	-0.02	-0.03	-0.03	-0.03
INR	0.02	-0.01	0.00	0.05	NA
Bilirubin subtype by gender					
Bilirubin/female	0.05	0.04	0.00	0.01	NA
Bilirubin/male	0.05	0.01	0.03	0.10	NA
Creatinine subtype by gender					
Creatinine/female	-0.10	0.00	-0.06	-0.03	-0.03
Creatinine/male	-0.06	-0.03	-0.02	-0.03	NA
INR subtype by gender					
INR/female	-0.01	-0.01	-0.03	0.03	NA
INR/male	0.03	-0.01	0.02	0.07	NA
All subtypes/female	-0.01	0.01	-0.02	-0.02	-0.02
All subtypes/male	0.01	-0.01	0.01	0.02	NA

Negative numbers imply decreased survival probability and positive numbers imply increased survival probability. INR, international normalized ratio; ITT, intent-to-treat; MELD, model for end-stage liver disease.

Second, we found that AKI was associated with worse outcomes than CKD and AKI on CKD. Our study affirms recent findings that rather than a singular creatinine value, the pattern (CKD, AKI, or AKI on CKD) does matter.[8,22–24]

Our study has several implications. First, registrants with MELD scores driven by serum creatinine that improve on the waitlist should not necessarily be seen as being less sick.[25] This is particularly relevant in the era of availability of treatments for HRS-AKI like terlipressin that reduce serum creatinine without significantly altering survival. Alternatively, a subset of patients may have improved due to volume replacement or other treatment for AKI. Such patients may be disadvantaged since treatments reduce their MELD score and consequently, their priority for LT. Once a decision for LT has been made, one may need to continue to pursue other avenues to either maintain their priority for transplantation or improve their chances of receiving an organ (*e.g.* using donation after cardiac death or living donation). Second, further work needs to define the impact of chronic conditions and kidney trajectory on patient outcome. Though kidney pattern may simply be a surrogate for a sicker patient, even after adjusting for relevant patient characteristics, kidney subtype matters. Patients with AKI appeared to have a higher risk of poor outcomes. Kidney injury may be an early sign of prognosis and even if kidney injury improves, such patients are still in need of a transplant.

Table 3. Differences in outcomes by MELD subtype for ITT survival, 1 year survival post-liver transplant, waitlist mortality and liver transplant rates within 90 days of registration.

MELD subtype	All MELD ranges	MELD <15	15≤ MELD <25	25≤ MELD <35	MELD ≥35
Overall ITT survival at 1 year					
Bilirubin	0.78	0.86	0.76	0.58	na
Creatinine	**0.65**	**0.82**	**0.71**	**0.50**	**0.23**
INR	0.75	0.83	0.74	0.58	na
All subtypes	0.73	0.84	0.74	0.52	0.27
p value	<0.001	<0.001	0.002	0.100	<0.001
Post-liver transplant survival at 1 year (subset that are transplanted)					
Bilirubin	0.94	0.95	0.95	0.94	na
Creatinine	**0.90**	**0.91**	**0.92**	**0.90**	**0.90**
INR	0.94	0.94	0.95	0.94	na
All subtypes	0.93	0.94	0.94	0.93	0.91
p value	<0.001	0.002	0.001	<0.001	=0.001
Waitlist mortality at 90 days					
Bilirubin	0.06	0.03	0.06	0.11	0.12
Creatinine	**0.12**	**0.04**	**0.07**	**0.15**	**0.22**
INR	0.08	0.03	0.06	0.10	0.22
All subtypes	0.08	0.03	0.06	0.12	0.21
p value	<0.001	<0.001	<0.001	<0.001	<0.001
Liver transplant rates at 90 days					
Bilirubin	0.40	0.15	0.38	0.77	0.85
Creatinine	**0.44**	**0.14**	**0.32**	**0.50**	**0.74**
INR	0.44	0.15	0.40	0.78	0.76
All subtypes	0.43	0.15	0.37	0.68	0.76
p value	<0.001	<0.001	<0.001	<0.001	<0.001

INR, international normalized ratio; ITT, intent-to-treat; MELD, model for end-stage liver disease.
Log-rank tests were used to compare the survival probability among MELD subtype groups.
The p-values for waitlist mortality at 90 days and liver transplant rates at 90 days come from comparing equality of the subdistribution for MELD subtype groups in cumulative incidence analysis.
Bold highlights the MELD Creatinine subtype.

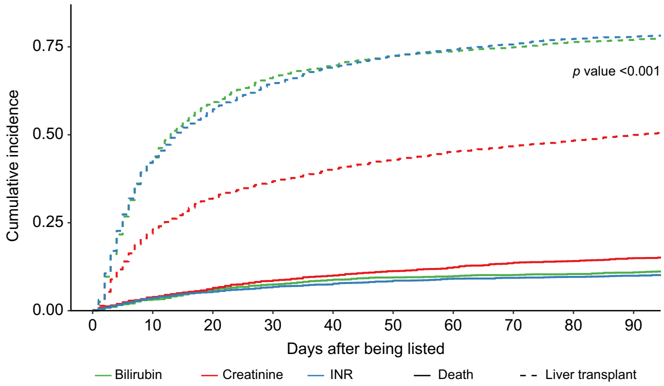


Fig. 3. Variation in liver transplantation rates by MELD subtype after accounting for competing risk of death on the waitlist for MELD scores 25-34 using Fine-Gray analysis. Solid line: death rates; Dashed lines: liver transplant rates. INR, international normalized ratio; MELD, model for end-stage liver disease. The *p*-value comes from comparing equality of the subdistribution for MELD subtype groups in cumulative incidence analysis. (This figure appears in color on the web.)

Third, our findings have relevance for the patient-physician discussion regarding risk of mortality. Our historical emphasis on 3-month mortality may need to be modified in our conversations to be more nuanced and impacted by MELD subtype patterns. Though MELD, and other post survival models, are suboptimal predictors of post-transplant outcomes, MELD-Cr was associated with the least favorable outcomes[26] Such honest discussion may allow for patients with limited choices beyond deceased donation to appropriately understand their relevant risk of untoward outcomes after waitlisting.

Our study continues to highlight disparities in LT rates for women. A lower LT rate and ITT survival was seen across almost all MELD subtypes and MELD strata. Recent work to incorporate female sex is encouraging (MELD 3.0, MELD GRAIL). However, drivers of mortality across different subsets remain less well characterized. For example, in the MELD-Br subtype, lower ITT survival may be driven by the presence of comorbidities or sarcopenia. In MELD-Cr, differences may be exacerbated by incomplete assessment of true kidney dysfunction, especially as it relates to muscle mass, age, and sex. Focused study outside the scope of this investigation will need to identify local drivers that may help improve ITT survival for women. Future studies should also investigate how allocation based on MELD 3.0 and continuous distribution models in the US may affect the patterns observed herein.

Our study has several strengths. Using statistical analysis techniques, we were able to classify patients into distinct phenotypes with relevant outcomes. We were able to examine the impact of sex, sodium, as well as relevant patient characteristics on explaining some of the differences by MELD subtype in a large database where data are collected in a protocolized fashion depending on MELD scores. We were able to examine relevant clinical drivers of the difference that capture the patient journey on the waitlist (change in MELD, type of kidney dysfunction and access to dual organ transplantation). However, our study has some limitations. We did not examine change in MELD subtype over time, and whether fluctuations from one subtype or another may impact outcomes. In addition, we did not examine whether center variation may mitigate some of the patient-level drivers of differences by subtypes. Listing practices as well as temporary inactivation of patient registration while decisions regarding progression of kidney function are being made vary by center. In the future, studying why certain centers are more adept and mitigating subtype differences may be helpful to improve ITT survival across the system. Definitions of CKD using registry data may be inaccurate. Patients may be misclassified as having CKD with incomplete information about baseline kidney function within 3 months prior to registration. In addition, we were unable to assess functional *vs.* structural AKI and their impact on prognosis. Certain conditions such as HRS-AKI may benefit from priority but this was not studied.[27]

As we move to consideration of continuous distribution, start evaluating utility and urgency in our patient selection, and expand the organ pool beyond deceased brain death donors, our study highlights MELD subtype as a potentially important driver. If our

Table 4. The association of MELD-creatinine with waitlist mortality and liver transplant rates.

	Waitlist mortality			Liver transplant		
	HR	95% CI	*p* value	HR	95% CI	*p* value
Age	1.02	1.02–1.02	0.000	1.00	1.00–1.00	<0.001
Female	1.10	1.04–1.16	0.002	0.91	0.88–0.95	<0.001
ALD	0.96	0.90–1.02	0.184	0.95	0.91–0.99	0.019
Cholestatic	0.82	0.74–0.90	<0.001	1.11	1.05–1.17	<0.001
HCV	1.06	0.97–1.14	0.193	0.97	0.92–1.03	0.356
NASH/cryptogenic	0.90	0.84–0.96	0.003	1.10	1.05–1.16	<0.001
Race (non-white)	1.09	1.01–1.16	0.022	0.91	0.87–0.96	<0.001
Latino	1.19	1.12–1.26	<0.001	0.86	0.82–0.89	<0.001
MELD-Na	0.99	0.99–1.00	<0.001	1.08	1.07–1.08	0.000
Hyponatremia	0.97	0.92–1.01	0.162	1.05	1.02–1.08	0.002
Weight	1.00	1.00–1.00	0.040	1.00	1.00–1.00	0.059
Height	0.99	0.99–1.00	<0.001	1.01	1.00–1.01	<0.001
BMI	1.00	1.00–1.00	0.809	1.00	1.00–1.00	0.404
Ventilator	2.10	1.87–2.37	0.000	0.61	0.55–0.67	0.000
MELD-creatinine	1.34	1.28–1.40	0.000	0.69	0.66–0.71	0.000

ALD, alcohol-related liver disease; HR, hazard ratio; MELD, model for end-stage liver disease; NASH, non-alcoholic steatohepatitis.
For the competing risk analysis, Fine-Gray proportional subdistribution hazards regression models were fitted.
The *p*-value for each variable comes from competing risk regression models for liver transplant and waitlist mortality.

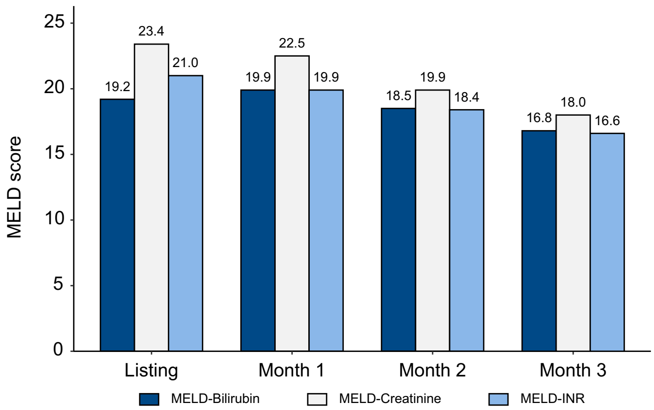


Fig. 4. MELD scores at 1, 2, and 3 months after listing by MELD subtypes. INR, international normalized ratio; MELD, model for end-stage liver disease.

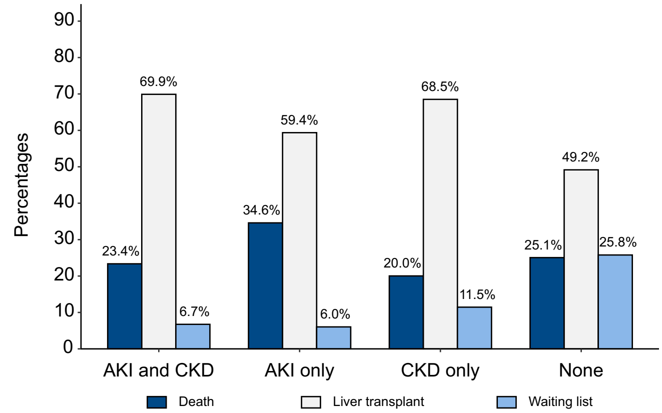


Fig. 5. Distribution of clinical outcomes in the MELD-Cr subtype by kidney dysfunction phenotypes. AKI, acute kidney injury; CKD, chronic kidney disease; MELD, model for end-stage liver disease.

findings are confirmed by other investigators in the non-US setting, it may provide credence to identifying a novel high-risk group for untoward outcomes and extra attention. Specifically, in the era of novel therapeutics (for example, terlipressin in the US), it may provide some support to the idea of assigning organ allocation priority using the highest serum creatinine rather than the post-treatment improvement in creatinine in patients with HRS.[28] Kidney sparing strategies for waitlisted patients are a matter of urgent priority. Further independent assessment and study of relevant implications on morbidity after transplant and improving LT rates for women is highly encouraged.

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Abbreviations

AKI, acute kidney injury; CKD, chronic kidney injury; HRS, hepatorenal syndrome; INR, international normalized ratio; ITT, intent-to-treat; LT, liver transplant; MELD, model for end-stage liver disease; bilirubin dominant MELD; MELD-Cr, creatinine dominant; MELD-INR, INR dominant MELD; SLKT, simultaneous liver-kidney transplant; SRTR, Scientific Registry of Transplant Recipients.

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Conflicts of interest

No personal or financial conflict of interest for any of the authors.

Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

Authors had access to all the study data, take responsibility for the accuracy of the analysis, and had authority over manuscript preparation and the decision to submit the manuscript for publication. All authors approve the manuscript.

Data availability statement

This study used data from the Scientific Registry of Transplant Recipients (SRTR). The SRTR data system includes data on all donor, wait-listed candidates, and transplant recipients in the United States (US) submitted by the members of the Organ Procurement and Transplantation Network (OPTN).

Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhep.2024.08.006.

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Author names in bold designate shared co-first authorship

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